Part II - PROJECT DESCRIPTION

Project Title

EduGuide PH: An AI-Powered Study and Career Guidance System

The proponents chose this title because it clearly describes the system's purpose and core functionality. The name "EduGuide PH" reflects its role as an educational guide specifically designed for the Philippine context, while "AI-Powered Study and Career Guidance System" highlights its key features: artificial intelligence for academic support and a dedicated focus on career pathing. The title emphasizes the system's goal of providing personalized, accessible, and intelligent guidance to students.

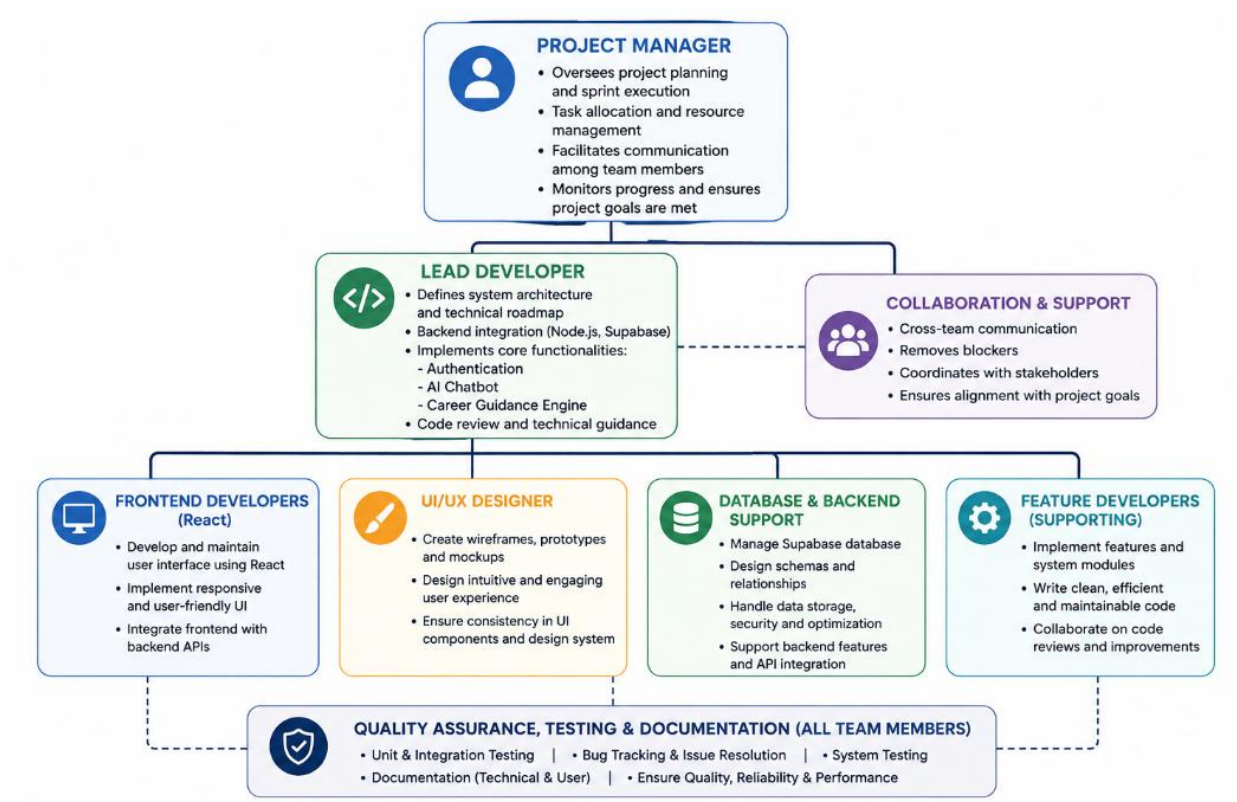
Project Logo

Project Organization

The project is led by the Project Manager, who oversees project planning, sprint execution, task allocation, and communication among team members. The Lead Developer is responsible for the system architecture, backend integration (Node.js, Supabase), and implementation of core functionalities such as authentication, the AI chatbot, and the career guidance engine. Supporting developers focus on frontend development (React), UI/UX design, database management, and feature implementation. Team members also collaborate on testing, documentation, and quality assurance activities to ensure the system meets project requirements.

The organizational structure promotes clear communication, accountability, and efficient coordination among members while maintaining flexibility for collaborative problem-solving and development.

The development of EduGuide PH follows the Agile Scrum Methodology, which emphasizes iterative development, collaboration, continuous feedback, and incremental delivery of system features. The project is divided into multiple activities performed across sprints, allowing the team to develop, test, and refine the system based on project requirements and stakeholder feedback.



Project Methodology

Phase 1 – Requirements Gathering and Planning

The team gathers user requirements, analyzes existing problems (as detailed in the PIECES analysis), and conducts meetings with stakeholders (school administrators, teachers, guidance counselors, and students) to define project objectives, scope, and system requirements.

Phase 2 – System Development

The team develops the core functionalities of EduGuide PH, including user authentication, the AI chatbot, career guidance engine, multilingual support, and database integration. Features are implemented incrementally throughout the development sprints.

Phase 3 – Testing and Enhancement

The system undergoes functional and integration testing to identify defects and verify that requirements are met. Necessary improvements and refinements are implemented based on testing results and user feedback.

Phase 4 – Deployment and Implementation

The completed system is prepared for deployment and implementation. Final validation, documentation, and system handover activities are conducted to ensure readiness for use at Amigo School of Calinan.

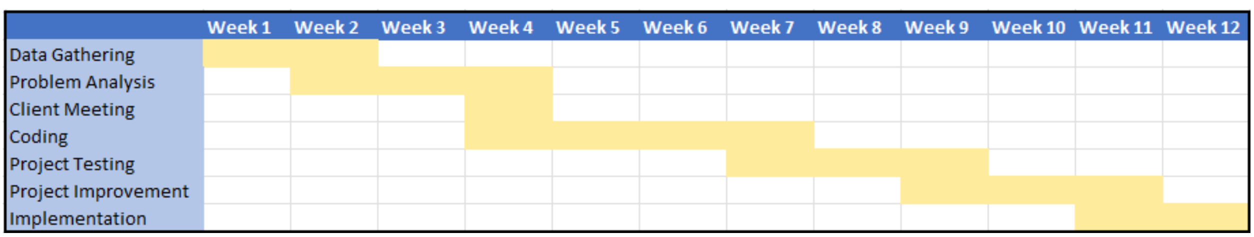
Project Schedule

Table 2.1 Project Schedule

Activity / Task	Timeline	Person Assigned	Expected Deliverable
Phase 1 – Requirements Gathering and Planning			
Data Gathering	Week 1–2	All Members	Requirements specification document, interview notes, user requirements
Problem Analysis	Week 2–4	All Members	PIECES Matrix, Problem-Opportunity Matrix (POM), system requirements analysis
Client Meeting	Week 4	All Members	Approved project scope, validated requirements, meeting minutes
Phase 2 – System Development			
Coding	Week 4–8	Development Team	Functional EduGuide PH web application modules, integrated features, source code repository
Phase 3 – Testing and Enhancement			

Project Testing	Week 8–10	All Members	Test cases, bug reports, testing documentation, validated system functions
Project Improvement	Week 10–11	Development Team	Resolved issues, optimized features, revised application version
Phase 4 – Deployment and Implementation			
Implementation	Week 11–12	All Members	Deployed EduGuide PH application, user documentation, final project deliverables

Gantt Chart



Technology

EduGuide PH is developed using a modern web-based and cloud-based technology stack that supports AI-powered academic support, real-time interactions, and secure data management. The tables below show the technology used:

Table 2.2 Technology Stack of EduGuide PH

Technology Category	Technology Used	Purpose
Programming Language	JavaScript, TypeScript	Used for implementing the application's functionality and business logic.
Frontend Framework	React	Enables the development of a dynamic, responsive, and user-friendly web interface.
Backend Framework	Node.js	Handles server-side logic, API development, and database interactions.
Database Management System	Supabase PostgreSQL	Stores user accounts, student interactions, query logs, quiz data, and career information.
Authentication Service	Supabase Authentication	Provides secure user registration and login functionality.
AI Integration	Google Gemini API	Provides the core AI capabilities for the chatbot, and career path suggestions.
Hosting Platform	Vercel	Used for deploying and hosting the web application.
IDE / Code Editor	VS Code / Cursor	Used for code development, debugging, and project management.
Version Control System	Git	Tracks and manages source code changes.
Repository Hosting Platform	GitHub	Supports source code storage, collaboration, and version management.
Testing Platform	Web Browsers (Chrome, Edge)	Used for application testing and validation.

Internet Connectivity	Broadband Internet Connection	Enables access to cloud services, APIs, and online resources.
Device	Laptop(s)	Used for developing and testing the EduGuide PH application.